



Additional Practice Questions for Chapter 8A: Integration Techniques

1. Integrate the following with respect to x .

- (a) $\left(1 - \frac{1}{x}\right)^2$ (b) $x \sin 3x$ (c) $\frac{x^2}{e^x}$ (d) $\frac{x}{1+x}$
 (e) $3 \ln x^2$ (f) $\sin x \cos^4 x$ (g) $\frac{x}{9x^2+1}$ (h) $\frac{1}{(4-5x)}$
 (i) $\frac{1}{(4-5x)^2}$ (j) $\sin^2 2x$ (k) $\frac{x}{x^4+4}$ (l) $\frac{x}{x^4-4}$
 (m) $\tan 2x \sec 2x$ (n) $\frac{(x+1)^2}{x^2+1}$ (o) $\cot^2 3x$ (p) $(e^x + x)^2$
 (q) $\frac{1}{x^2(x-1)}$ (r) $\frac{1}{\sqrt{1-2x^2}}$ (s) $\frac{x}{\sqrt{1-2x^2}}$ (t) $x \cos\left(\frac{x^2+1}{2}\right)$
 (u) $\frac{3x^2+x+4}{x^3+x}$

(a) $\int \left(1 - \frac{1}{x}\right)^2 dx$ $= \int 1 - \frac{2}{x} + \frac{1}{x^2} dx$ $= x - 2 \ln x - \frac{1}{x} + c$	(l) $\int \frac{x}{x^4-4} dx$ $= \frac{1}{2} \int \frac{2x}{(x^2)^2-4} dx$ $= \frac{1}{8} \ln \left \frac{x^2-2}{x^2+2} \right + c$
(b) $\int x \sin 3x dx$ $= \frac{-\cos 3x}{3} \cdot x - \int \frac{-\cos 3x}{3} dx$ $= \frac{-x \cos 3x}{3} + \frac{\sin 3x}{9} + c$	(m) $\int \tan 2x \sec 2x dx$ $= \frac{1}{2} \sec 2x + c$
(c) $\int \frac{x^2}{e^x} dx$ $= (-e^{-x})x^2 - \int (-e^{-x})(2x) dx$ $= -x^2 e^{-x} + (-e^{-x})(2x) - \int (-e^{-x})(2) dx$ $= -e^{-x}(x^2 + 2x + 2) + c$	(n) $\int \frac{(x+1)^2}{x^2+1} dx$ $= \int 1 + \frac{2x}{x^2+1} dx$ $= x + \ln(x^2+1) + c$

(d) $\int \frac{x}{1+x} dx$ $= \int 1 - \frac{1}{1+x} dx$ $= x - \ln 1+x + c$	(o) $\int \cot^2 3x dx$ $= \int \operatorname{cosec}^2 3x - 1 dx$ $= -\frac{1}{3} \cot 3x - x + c$
(e) For $x > 0$, $\int 3 \ln x^2 dx = \int 6 \ln x dx$ $= 6 \left[x \ln x - \int 1 dx \right]$ $= 6x(\ln x - 1) + c$ For $x < 0$, $\int 3 \ln x^2 dx = \int 6 \ln(-x) dx$ (NB: $\ln x$ is undefined since $x < 0$) $= 6 \left[x \ln(-x) - \int \frac{-1}{-x} x dx \right]$ $= 6x[\ln(-x) - 1] + c$	(p) $\int (e^x + x)^2 dx$ $= \int e^{2x} + 2xe^x + x^2 dx$ $= \frac{e^{2x}}{2} + 2xe^x - 2e^x + \frac{x^3}{3} + c$
(f) $\int \sin x \cos^4 x dx$ $= -\frac{\cos^5 x}{5} + c$	(q) $\int \frac{1}{x^2(x-1)} dx$ $= \int -\frac{1}{x^2} - \frac{1}{x} + \frac{1}{x-1} dx$ $= \frac{1}{x} - \ln x + \ln x-1 + c$
(g) $\int \frac{x}{9x^2+1} dx$ $= \frac{1}{18} \ln(9x^2+1) + c$	(r) $\int \frac{1}{\sqrt{1-2x^2}} dx$ $= \frac{1}{\sqrt{2}} \sin^{-1}(\sqrt{2}x) + c$
(h) $\int \frac{1}{4-5x} dx$ $= -\frac{1}{5} \ln 4-5x + c$	(s) $\int \frac{x}{\sqrt{1-2x^2}} dx$ $= -\frac{1}{4} \frac{\sqrt{1-2x^2}}{1/2} + c$ $= -\frac{\sqrt{1-2x^2}}{2} + c$
(i) $\int \frac{1}{(4-5x)^2} dx$ $= \frac{1}{5(4-5x)} + c$	(t) $\int x \cos\left(\frac{x^2+1}{2}\right) dx$ $= \sin\left(\frac{x^2+1}{2}\right) + c$

(j) $\int \sin^2 2x \, dx$ $= \int \frac{1 - \cos 4x}{2} \, dx$ $= \frac{x}{2} - \frac{\sin 4x}{8} + c$	(u) $\int \frac{3x^2 + x + 4}{x^3 + x} \, dx$ $= \int \frac{3x^2 + x + 4}{x(x^2 + 1)} \, dx$ $= \int \frac{4}{x} + \frac{1-x}{x^2 + 1} \, dx$ $= \int \frac{4}{x} + \frac{1}{x^2 + 1} - \frac{1}{2} \cdot \frac{2x}{x^2 + 1} \, dx$ $= 4 \ln x + \tan^{-1} x - \frac{1}{2} \ln(x^2 + 1) + c$
(k) $\int \frac{x}{x^4 + 4} \, dx$ $= \frac{1}{2} \int \frac{2x}{(x^2)^2 + 4} \, dx$ $= \frac{1}{4} \tan^{-1} \left(\frac{x^2}{2} \right) + c$	

2. By expressing $\cos \theta - \sqrt{3} \sin \theta$ as $R \cos(\theta + \alpha)$, where R and α are constants to be determined, find $\int \frac{1}{\cos \theta - \sqrt{3} \sin \theta} \, d\theta$. [2]

2. $\cos \theta - \sqrt{3} \sin \theta = 2 \cos\left(\theta + \frac{\pi}{3}\right)$

$$\int \frac{1}{\cos \theta - \sqrt{3} \sin \theta} \, d\theta$$

$$= \int \frac{1}{2} \sec\left(\theta + \frac{\pi}{3}\right) \, d\theta$$

$$= \frac{1}{2} \ln \left| \sec\left(\theta + \frac{\pi}{3}\right) + \tan\left(\theta + \frac{\pi}{3}\right) \right| + c_{\#}$$

3. (a) $\int x^2 \cos 2x \, dx$ (b) $\int x\sqrt{x^2+1} \, dx$ (c) $\int x \sin x^2 \, dx$
 (d) $\int \frac{2}{\sqrt{9-4x^2}} \, dx$ (e) $\int \frac{x}{(x+1)^2} \, dx$ (f) $\int x^3 e^{2x} \, dx$

3a) $\int x^2 \cos 2x \, dx$ $= \frac{\sin 2x}{2} (x^2) - \int \frac{\sin 2x}{2} (2x) \, dx$ $= \frac{x^2 \sin 2x}{2} + \frac{x \cos 2x}{2} - \int \frac{\cos 2x}{2} \, dx$ $= \frac{x^2 \sin 2x}{2} + \frac{x \cos 2x}{2} - \frac{\sin 2x}{4} + c_{\#}$	b) $\int x\sqrt{x^2+1} \, dx$ $= \frac{1}{2} \frac{(x^2+1)^{3/2}}{3/2} + c$ $= \frac{(x^2+1)^{3/2}}{3} + c_{\#}$
c) $\int x \sin x^2 \, dx = -\frac{1}{2} \cos x^2 + c_{\#}$	d) $\int \frac{2}{\sqrt{9-4x^2}} \, dx = \sin^{-1} \left(\frac{2x}{3} \right) + c_{\#}$
e) $\int \frac{x}{(x+1)^2} \, dx$ $= \int \frac{1}{x+1} - \frac{1}{(x+1)^2} \, dx$ $= \ln x+1 + \frac{1}{x+1} + c_{\#}$	f) $\int x^3 e^{2x} \, dx$ $= \frac{e^{2x}}{2} x^3 - \int \frac{e^{2x}}{2} 3x^2 \, dx$ $= \frac{e^{2x}}{2} x^3 - \frac{3}{2} \frac{e^{2x}}{2} x^2 + \int \frac{3}{2} \frac{e^{2x}}{2} 2x \, dx$ $= e^{2x} \left(\frac{x^3}{2} - \frac{3x^2}{4} \right) + \frac{3}{2} \frac{e^{2x}}{2} x - \int \frac{3}{2} \frac{e^{2x}}{2} \, dx$ $= e^{2x} \left(\frac{x^3}{2} - \frac{3x^2}{4} + \frac{3x}{4} - \frac{3}{8} \right) + c_{\#}$

4. By using the identity $\sin^2 x + \cos^2 x = 1$, determine $\int \sin^3 x \, dx$.

4. $\int \sin^3 x \, dx$

$$= \int (1 - \cos^2 x) \sin x \, dx$$

$$= -\cos x + \frac{\cos^3 x}{3} + c$$

5. By using the given substitution, determine the following integrals, leaving your answers in the exact form whenever applicable.

(a) $\int \tan \frac{x}{2} \sec^4 \frac{x}{2} dx$ $[u = \sec \frac{x}{2}]$	(b) $\int_1^e \frac{1}{y\sqrt{1-(\ln y)^2}} dy$ $[u = \ln y]$
(c) $\int_0^1 \frac{1}{(4-t^2)^{\frac{3}{2}}} dt$ $[t = 2 \sin \theta]$	(d) $\int_2^3 \sqrt{\frac{x-2}{4-x}} dx$ $[x = 2 + 2 \cos^2 \theta]$

Note: Try solving (a) and (b) without using substitution.

5(a) $\int \tan \frac{x}{2} \sec^4 \frac{x}{2} dx$ $= \int 2u^3 du$ $= \frac{u^4}{2} + c$ $= \frac{1}{2} \sec^4 \frac{x}{2} + c$ 5(a) Alternatively, $\int \tan \frac{x}{2} \sec^4 \frac{x}{2} dx$ $= \int \tan \frac{x}{2} \sec \frac{x}{2} \sec^3 \frac{x}{2} dx$ $= \frac{1}{2} \sec^4 \frac{x}{2} + c$	5(b) $\int_1^e \frac{1}{y\sqrt{1-(\ln y)^2}} dy$ $= \int_0^1 \frac{1}{\sqrt{1-u^2}} du$ $= [\sin^{-1} u]_0^1$ $= \frac{\pi}{2}$ Alternatively, $\int_1^e \frac{1}{y\sqrt{1-(\ln y)^2}} dy$ $= [\sin^{-1}(\ln y)]_1^e$ $= \sin^{-1}(1) = \frac{\pi}{2}$
5(c) $\int_0^1 \frac{1}{(4-t^2)^{\frac{3}{2}}} dt$ $= \int_0^{\frac{\pi}{6}} \frac{1}{(4-4\sin^2 \theta)^{\frac{3}{2}}} 2\cos \theta d\theta$ $= \int_0^{\frac{\pi}{6}} \frac{1}{(4\cos^2 \theta)^{\frac{3}{2}}} 2\cos \theta d\theta$ $= \int_0^{\frac{\pi}{6}} \frac{1}{4} \sec^2 \theta d\theta$ $= \frac{1}{4} [\tan \theta]_0^{\frac{\pi}{6}}$ $= \frac{1}{4\sqrt{3}} = \frac{\sqrt{3}}{12}$	5(d) $\int_2^3 \sqrt{\frac{x-2}{4-x}} dx$ $= \int_{\frac{\pi}{2}}^{\frac{\pi}{4}} \sqrt{\frac{2\cos^2 \theta}{2-2\cos^2 \theta}} \cdot (-4\cos \theta \sin \theta) d\theta$ $= 4 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sqrt{\frac{\cos^2 \theta}{\sin^2 \theta}} \cos \theta \sin \theta d\theta$ $= 4 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^2 \theta d\theta$ $= 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos 2\theta + 1 d\theta$ $= 2 \left[\frac{\sin 2\theta}{2} + \theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} = \frac{\pi}{2} - 1$

6. Determine the exact value of $\int_0^1 \frac{x+9}{2x^2+x-6} dx$.

$$\begin{aligned} 6. & \int_0^1 \frac{x+9}{2x^2+x-6} dx \\ &= \int_0^1 \frac{x+9}{(2x-3)(x+2)} dx \\ &= \int_0^1 \frac{3}{2x-3} - \frac{1}{x+2} dx \\ &= \left[\frac{3}{2} \ln|2x-3| - \ln|x+2| \right]_0^1 \\ &= -\ln 3 - \frac{3}{2} \ln 3 + \ln 2 = \ln 2 - \frac{5}{2} \ln 3 \end{aligned}$$

7. By the use of the substitution $x = 3 \sin t$, determine the exact value of $\int_0^3 x^2 \sqrt{9-x^2} dx$.

$$\begin{aligned} 7. & \int_0^3 x^2 \sqrt{9-x^2} dx \\ &= \int_0^{\frac{\pi}{2}} 9 \sin^2 t \sqrt{9-9 \sin^2 t} \cdot 3 \cos t dt \\ &= \int_0^{\frac{\pi}{2}} 81 \sin^2 t \cos^2 t dt \\ &= \int_0^{\frac{\pi}{2}} \frac{81}{4} \sin^2 2t dt \\ &= \int_0^{\frac{\pi}{2}} \frac{81}{4} \cdot \frac{1-\cos 4t}{2} dt \\ &= \frac{81}{8} \left[t - \frac{\sin 4t}{4} \right]_0^{\frac{\pi}{2}} = \frac{81\pi}{16} \end{aligned}$$

8. VJC Prelim 9758/2017/01/Q9(a)

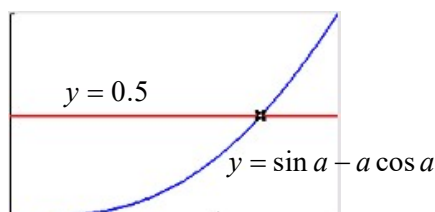
Given that $\int_0^a x \sin x \, dx = 0.5$, where $0 < a < 2$, find an equation that is satisfied by a and use it to find the value of a .

$$8. \int_0^a x \sin x \, dx = 0.5$$

$$[-x \cos x]_0^a + \int_0^a \cos x \, dx = 0.5$$

$$[-a \cos a + 0] + [\sin x]_0^a = 0.5$$

$$-a \cos a + \sin a = 0.5 \quad \text{--- (1)}$$



Using GC, $a = 1.20249 = 1.20$ (3 s.f.)

9. Express $\frac{x^2 + x + 28}{(1-x)(x^2 + 9)}$ in partial fractions.

Hence, show that $\int_0^3 \frac{x^2 + x + 28}{(1-x)(x^2 + 9)} \, dx = \frac{\pi}{12} - 2 \ln 2$.

9.

$$\frac{x^2 + x + 28}{(1-x)(x^2 + 9)} = \frac{C}{1-x} + \frac{Ax + B}{x^2 + 9}$$

$$x^2 + x + 28 = C(x^2 + 9) + (Ax + B)(1-x)$$

When $x = 1$, $C = 3$

$$x^2 + x + 28 = 3(x^2 + 9) + (Ax + B)(1-x)$$

When $x = 0$, $28 = 27 + B \Rightarrow B = 1$

Comparing coefficient of x^2 , $1 = 3 - A \Rightarrow A = 2$

$$\therefore \frac{x^2 + x + 28}{(1-x)(x^2 + 9)} = \frac{3}{1-x} + \frac{2x+1}{x^2 + 9}$$

$$\begin{aligned}
& \int_0^3 \frac{x^2 + x + 28}{(1-x)(x^2+9)} dx \\
&= \int_0^3 \frac{3}{1-x} + \frac{2x+1}{x^2+9} dx \\
&= \left[-3 \ln|1-x| + \ln|x^2+9| + \frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) \right]_0^3 \\
&= -3 \ln 2 + \ln 18 + \frac{\pi}{12} - \ln 9 \\
&= \ln\left(\frac{18}{9 \times 8}\right) + \frac{\pi}{12} \\
&= \frac{\pi}{12} + \ln \frac{1}{4} \\
&= \frac{\pi}{12} - 2 \ln 2_{\#} \text{ (Shown)}
\end{aligned}$$

10. By using the substitution $u = \sqrt{x+2}$, or otherwise, show that $\int_2^7 \frac{1}{(x+1)\sqrt{x+2}} dx = \ln \frac{3}{2}$.

Hence, find the exact value of $\int_2^7 \frac{(x-1)}{(x+1)\sqrt{x+2}} dx$.

$$\begin{aligned}
& \int_2^7 \frac{1}{(x+1)\sqrt{x+2}} dx \\
&= \int_2^3 \frac{1}{(u^2-1)u} 2u du \\
&= \left[\ln \left| \frac{u-1}{u+1} \right| \right]_2^3 \\
&= \ln \frac{2}{4} - \ln \frac{1}{3} \\
&= \ln \frac{3}{2}_{\#} \text{ (shown)}
\end{aligned}$$

$$\begin{aligned}
\text{Let } u &= \sqrt{x+2} \\
u^2 &= x+2 \\
\frac{dx}{du} &= 2u \\
\text{When } x &= 2, u = 2 \\
\text{When } x &= 7, u = 3
\end{aligned}$$

$$\begin{aligned}
 \int_2^7 \frac{(x-1)}{(x+1)\sqrt{x+2}} dx &= \int_2^7 \frac{(x+1)-2}{(x+1)\sqrt{x+2}} dx \\
 &= \int_2^7 \frac{1}{\sqrt{x+2}} - \frac{2}{(x+1)\sqrt{x+2}} dx \\
 &= \left[2\sqrt{x+2} \right]_2^7 - 2 \ln \frac{3}{2} \\
 &= 2 - 2 \ln \frac{3}{2}_{\#}
 \end{aligned}$$

11.

(a) $\int \frac{x+2}{\sqrt{1-3x^2}} dx$

(b) $\int \frac{2}{x \ln x^2} dx$

(c) $\int \frac{1}{1+e^x} dx$

(d) $\int \frac{1}{x[4+(\ln x)^2]} dx$

(e) $\int \ln(2x+1) dx$

(f) $\int \frac{6x+4}{(1-2x)(1+3x^2)} dx$

a) $\int \frac{x+2}{\sqrt{1-3x^2}} dx$ $ \begin{aligned} &= -\frac{1}{6} \int -6x(1-3x^2)^{-\frac{1}{2}} dx + 2 \int \frac{1}{\sqrt{1-3x^2}} dx \\ &= -\frac{1}{6} \frac{\sqrt{1-3x^2}}{\frac{1}{2}} + \frac{2}{\sqrt{3}} \sin^{-1}(\sqrt{3}x) + c \\ &= -\frac{1}{3} \sqrt{1-3x^2} + \frac{2}{\sqrt{3}} \sin^{-1}(\sqrt{3}x) + c_{\#} \end{aligned} $	b) $\int \frac{2}{x \ln x^2} dx = \ln \ln x^2 + c_{\#}$
c) $\int \frac{1}{1+e^x} dx = \int \frac{1+e^x - e^x}{1+e^x} dx$ $ \begin{aligned} &= \int 1 - \frac{e^x}{1+e^x} dx \\ &= x - \ln(1+e^x) + c_{\#} \end{aligned} $ or $\int \frac{1}{1+e^x} dx = \int \frac{e^{-x}}{e^{-x}(1+e^x)} dx$ $ \begin{aligned} &= -\int \frac{-e^{-x}}{1+e^{-x}} dx \\ &= -\ln(1+e^{-x}) + c_{\#} \end{aligned} $	d) $\int \frac{1}{x[4+(\ln x)^2]} dx$ $ = \frac{1}{2} \tan^{-1}\left(\frac{\ln x}{2}\right) + c_{\#} $

$$\begin{aligned}
 \text{e) } \int \ln(2x+1) \, dx &= x \ln(2x+1) - \int \frac{2x}{2x+1} \, dx \\
 &= x \ln(2x+1) - \int 1 - \frac{1}{2x+1} \, dx \\
 &= x \ln(2x+1) - x + \frac{1}{2} \ln(2x+1) + c_{\#}
 \end{aligned}$$

$$\begin{aligned}
 \text{f) } \int \frac{6x+4}{(1-2x)(1+3x^2)} \, dx &= \int \frac{4}{1-2x} + \frac{6x}{1+3x^2} \, dx \\
 &= -2 \ln|1-2x| + \ln(1+3x^2) + c_{\#}
 \end{aligned}$$

12. By using the substitution $u = x^4 + 1$, find $\int_1^3 \frac{1}{x(x^4+1)} \, dx$, leaving your answers in the exact form.

12.

$$\begin{aligned}
 &\int_1^3 \frac{1}{x(x^4+1)} \, dx \\
 &= \int_2^{82} \frac{1}{(u-1)^{1/4} u} \cdot \frac{1}{4(u-1)^{3/4}} \, du \\
 &= \int_2^{82} \frac{1}{4u(u-1)} \, du \\
 &= \frac{1}{4} \int_2^{82} \frac{-1}{u} + \frac{1}{(u-1)} \, du \\
 &= \frac{1}{4} \left[-\ln|u| + \ln|u-1| \right]_2^{82} \\
 &= \frac{1}{4} (-\ln 82 + \ln 81 + \ln 2) \\
 &= \frac{1}{4} \ln \left(\frac{81}{41} \right)_{\#}
 \end{aligned}$$

$$\text{Let } u = x^4 + 1 \Rightarrow x = (u-1)^{\frac{1}{4}}$$

$$\frac{dx}{du} = \frac{1}{4}(u-1)^{-\frac{3}{4}}$$

$$\text{When } x = 1, u = 1^4 + 1 = 2$$

$$\text{When } x = 3, u = 3^4 + 1 = 82$$

13. 9758/2019/02/Q1

You are given that $I = \int x(1-x)^{\frac{1}{2}} \, dx$.

- (i) Use integration by parts to find an expression for I . [2]
- (ii) Use the substitution $u^2 = 1-x$ to find another expression for I . [2]
- (iii) Show algebraically that your answers to part (i) and (ii) differ by a constant. [2]

$$\begin{aligned}
 \text{(i)} \quad I &= -\frac{2}{3}x(1-x)^{\frac{3}{2}} + \frac{2}{3} \int (1-x)^{\frac{3}{2}} dx \\
 &= -\frac{2}{3}x(1-x)^{\frac{3}{2}} - \frac{4}{15}(1-x)^{\frac{5}{2}} + c
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad x &= 1-u^2 \Rightarrow \frac{dx}{du} = -2u \\
 I &= \int (1-u^2) \sqrt{u^2} (-2u) du \\
 &= 2 \int (u^2 - 1) u^2 du \\
 &= 2 \int (u^4 - u^2) du \\
 &= \frac{2}{5}u^5 - \frac{2}{3}u^3 + d \\
 &= \frac{2}{5}(1-x)^{\frac{5}{2}} - \frac{2}{3}(1-x)^{\frac{3}{2}} + d
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad &\frac{2}{5}(1-x)^{\frac{5}{2}} - \frac{2}{3}(1-x)^{\frac{3}{2}} + d - \left(-\frac{2}{3}x(1-x)^{\frac{3}{2}} - \frac{4}{15}(1-x)^{\frac{5}{2}} + c \right) \\
 &= \frac{2}{3}x(1-x)^{\frac{3}{2}} - \frac{2}{3}(1-x)^{\frac{3}{2}} + \frac{4}{15}(1-x)^{\frac{5}{2}} + \frac{2}{5}(1-x)^{\frac{5}{2}} + d - c \\
 &= \frac{2}{3}(1-x)^{\frac{3}{2}}(x-1) + \frac{10}{15}(1-x)^{\frac{5}{2}} + d - c \\
 &= -\frac{2}{3}(1-x)^{\frac{3}{2}}(1-x) + \frac{2}{3}(1-x)^{\frac{5}{2}} + d - c \\
 &= -\frac{2}{3}(1-x)^{\frac{5}{2}} + \frac{2}{3}(1-x)^{\frac{5}{2}} + d - c \\
 &= d - c
 \end{aligned}$$